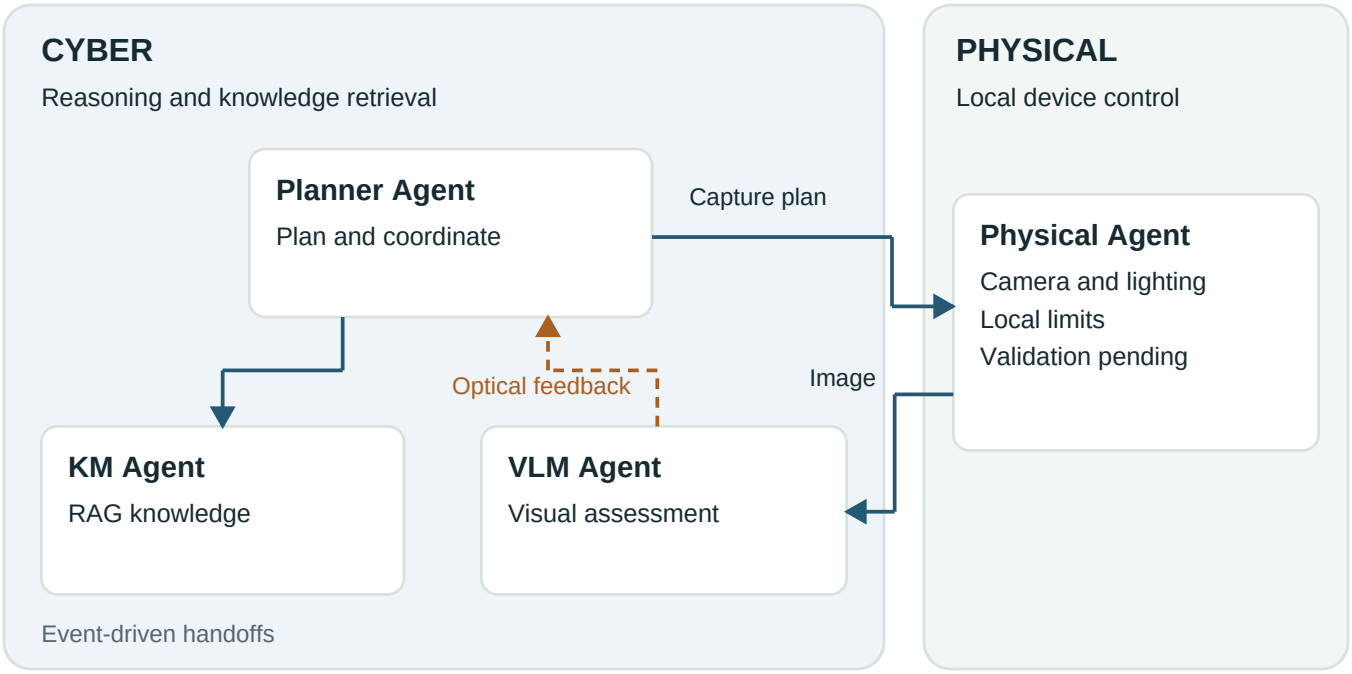
Server Assembly Inspection

Four-agent architecture and workflow design | In progress

I lead the overall architecture and agent workflow design for an in-progress server assembly inspection project. The design separates cloud reasoning and knowledge retrieval from local device execution and safety boundaries.



Simplified conceptual architecture based on the stated design. It is not a deployed topology or a hardware safety certification.

Inspection feedback loop

The Planner coordinates knowledge retrieval and a capture plan. The Physical Agent handles local camera and lighting actions. The VLM assesses the image; when reflection or shadow prevents a useful judgment, its optical diagnosis informs replanning and another capture.

Design decisions and validation

Confirmed design intent and proposed checks are kept distinct

Design decision	Reason and boundary
Cyber and Physical separation	Keep semantic planning and RAG in the cloud; keep device control and intended safety checks on the edge.
Specialist agents	Planner coordinates; KM retrieves knowledge; Physical executes locally; VLM assesses visual evidence.
Optical feedback	Distinguish a visibility problem from a confident inspection decision before requesting a new capture.
Local safety authority	A generated plan must not override the physical execution boundary. Actual safeguards still require implementation and validation.

Interface example

The following fields are a proposed synthetic message contract, not a claim about an implemented interface.

```
{
  "inspection_id": "demo-001",
  "event_type": "capture.review_requested",
  "plan_version": 1,
  "diagnosis": "visibility_limited",
  "optical_reason": "reflection",
  "next_action": "replan_capture"
}
```

Validation work to complete

- Validate knowledge retrieval and VLM judgments against a defined, approved inspection dataset.
- Test local action limits and feedback using an agreed hardware or simulation setup; do not infer collision safety from the diagram.
- Agree retry limits, timeout behavior, event correlation and human escalation, then test failure and repeated-event scenarios.

Status: architecture and workflow design in progress. No production deployment, inspection accuracy, completed four-agent implementation or safety performance is asserted.